



Lecture #2.4

ROS 2 and Machine Interface

MAS418

Programming for Intelligent Robotics and Industrial systems

Part II: PLC Software Development

Autumn 2025

Daniel Hagen, PhD

Previous Lecture

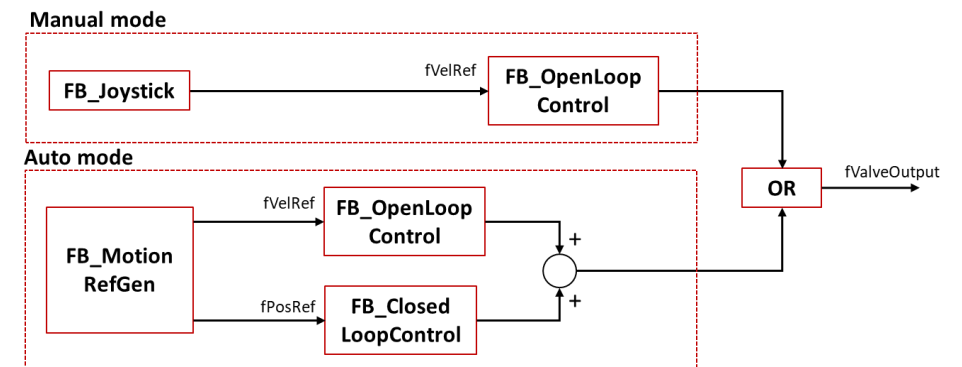
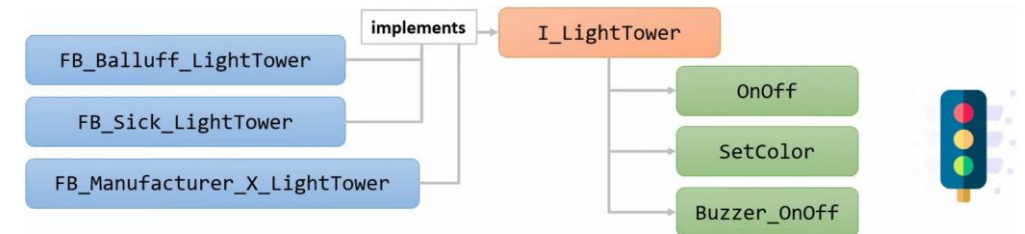
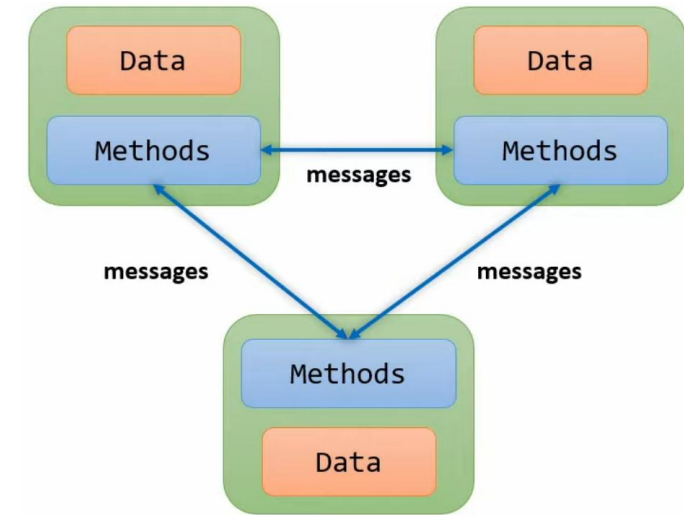
• Object-oriented PLC programming

I. Function Blocks

- Introduction
- Function blocks
- Methods
- Inheritance

II. Interfaces

III. Introduction to Exercise #2.3



Overview

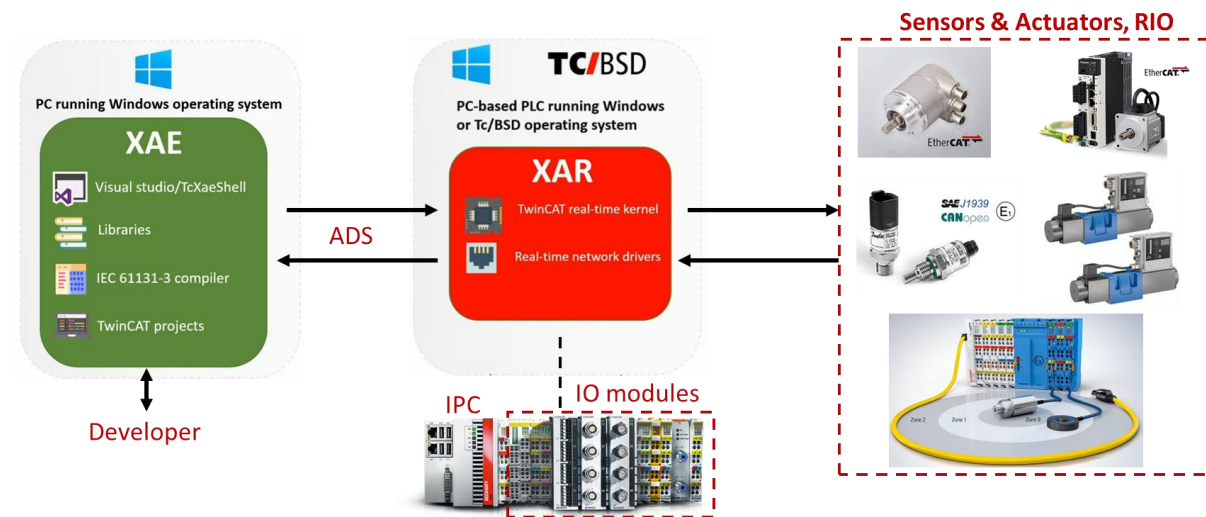
Introduction

Part I: Machine Interface

Part II: ROS 2 Interface

Part III: Demo

Summary

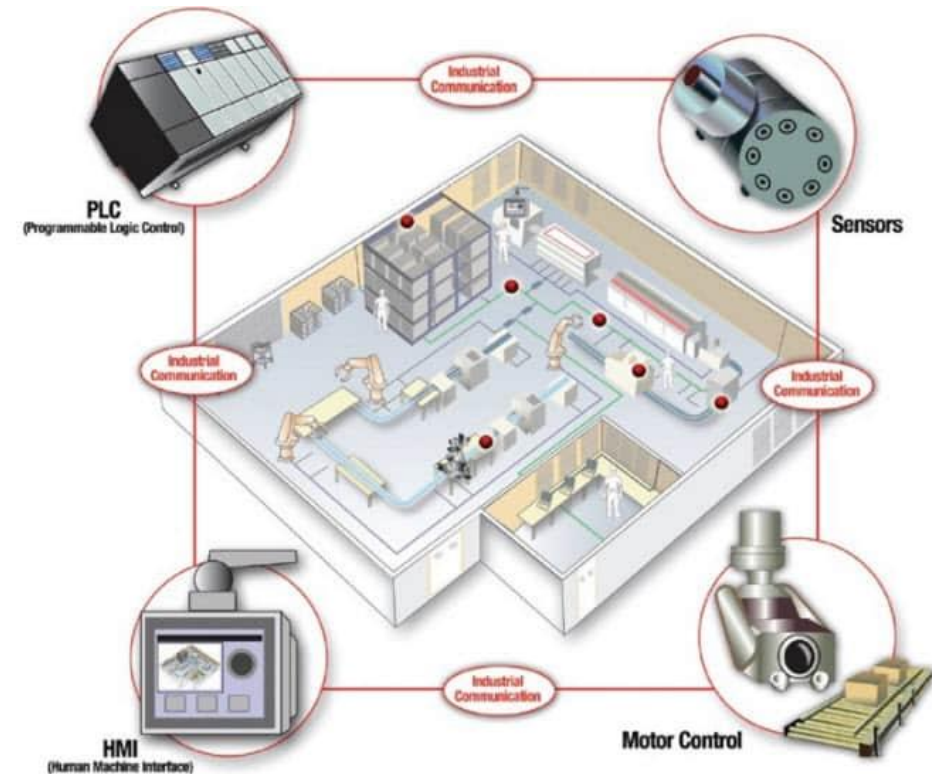


Part I: Machine Interface

1. Introduction
2. Input & Outputs (I/O)
3. Fieldbus
4. EtherCAT
5. I/O mapping

Introduction

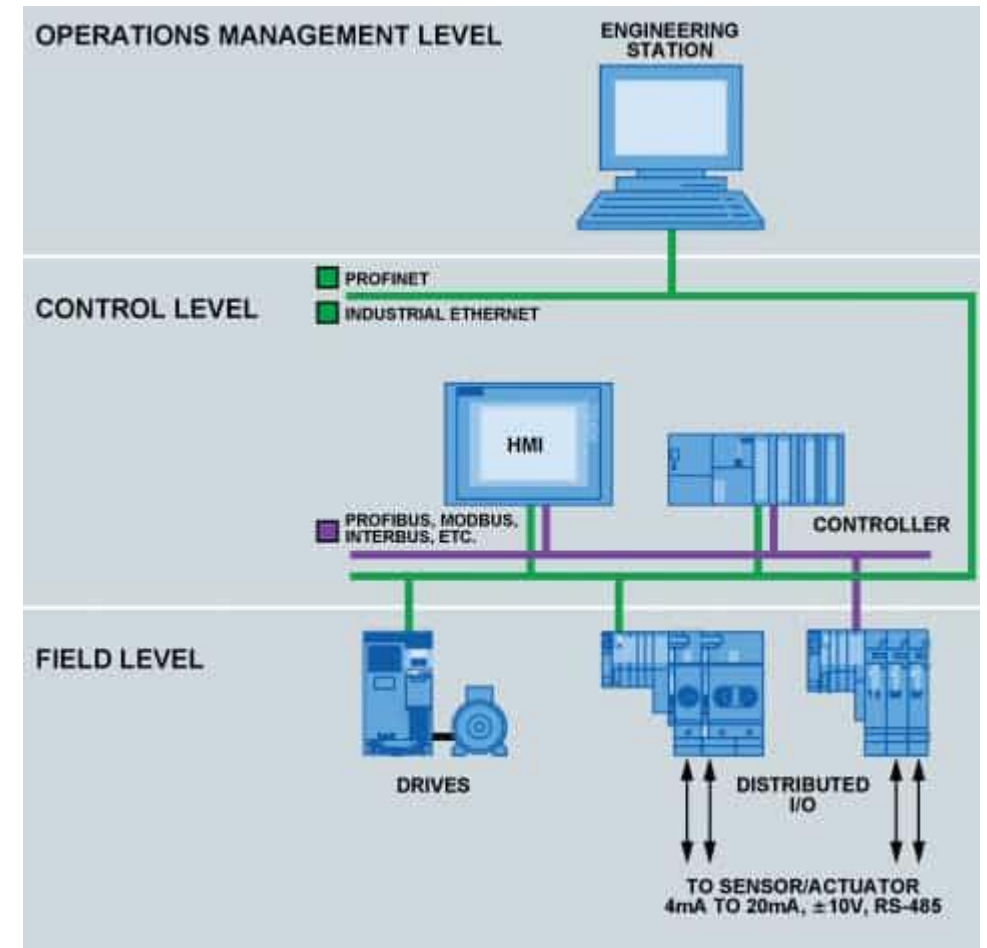
- An industrial network provides a powerful means of data exchange, data controllability, and flexibility to connect various devices
- Trend towards open systems interconnection standards that permits to interconnect and communicate any pair of automation devices reliably irrespective of the manufacturer
- Industrial network is a special type of network made to handle real-time control and data integrity in harsh environments over large installations
- With the advancements in digital technology, fieldbus technology is now ruling the automation field as it provides multidrop (master-slave configuration) communication facility that results cost effective and cable saving communication



Introduction

Hierarchical levels in industrial networks

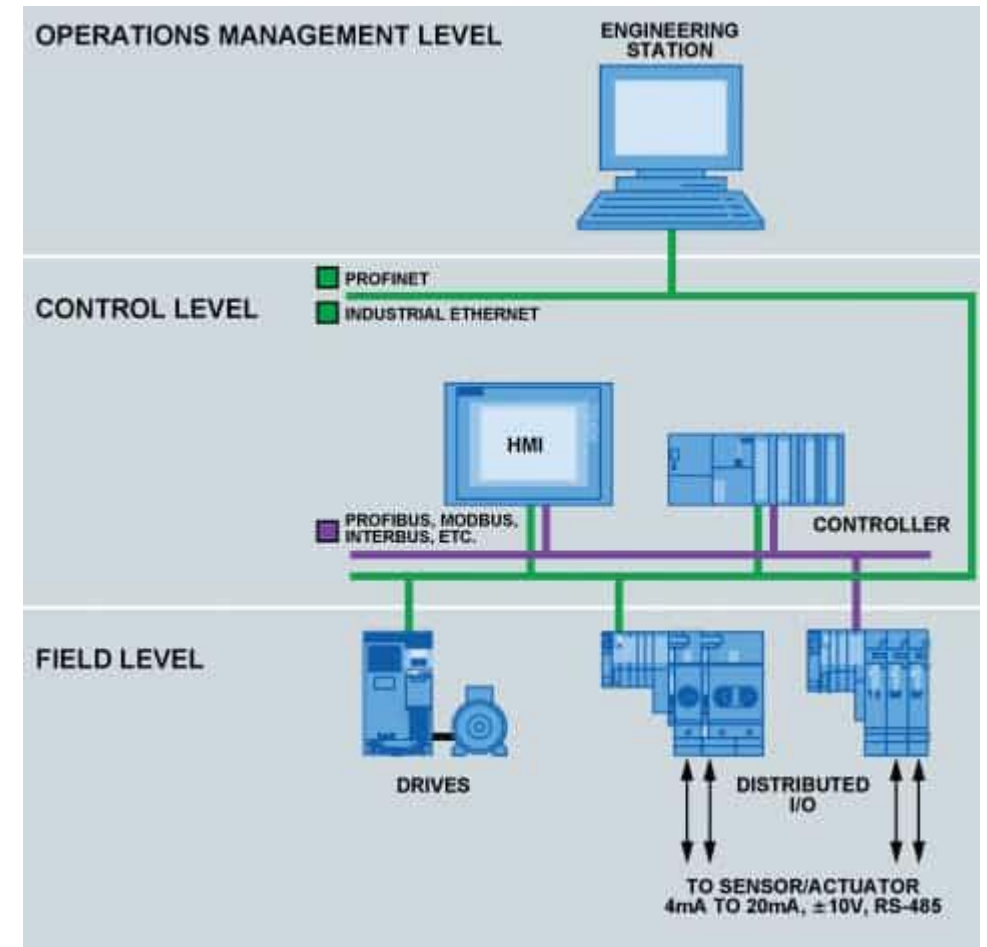
- In many industries, the information or data flows from field level to enterprise level (bottom-to-top) and vice-versa
- Different levels have to handle different requirements of a particular level
- No single communication network address requirements needed by each level
- Hence different levels may use different network based on the requirements such as data volume, data transmission, data security, etc.
- Based on the functionality, industrial communication networks are classified into three general levels



Introduction

Device/field level

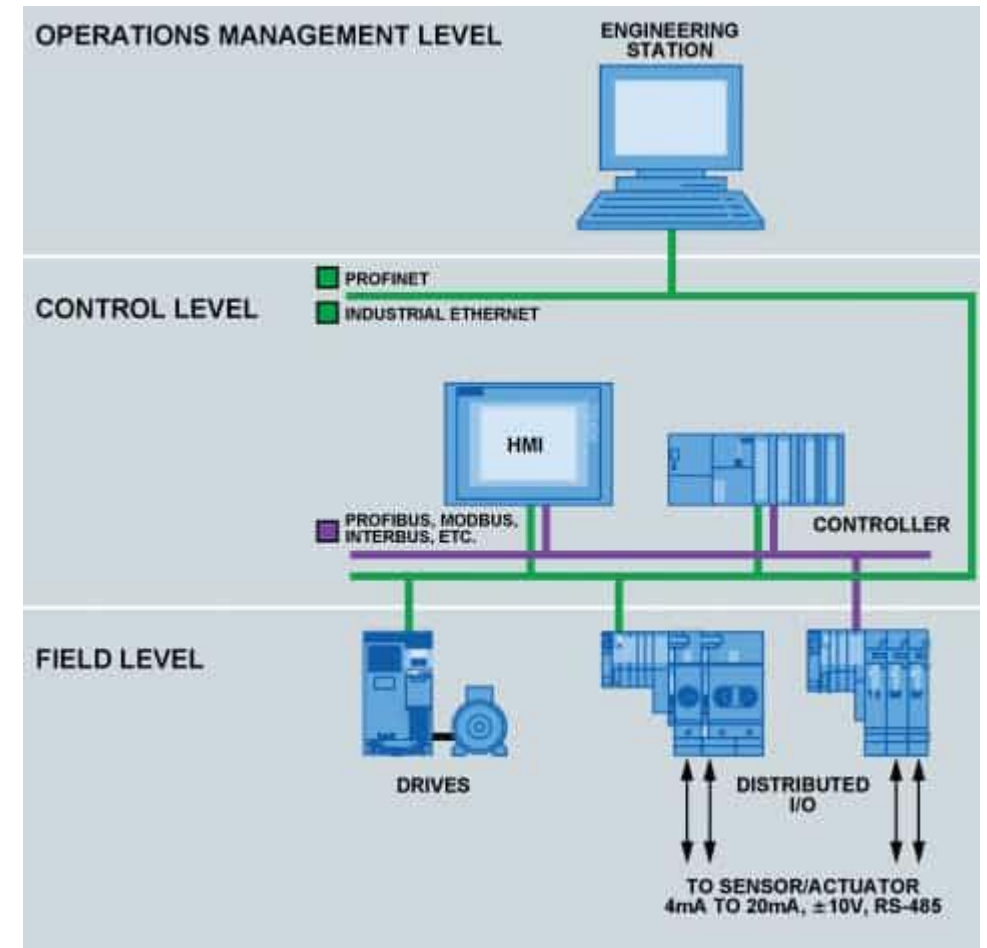
- This lowest level consists of field devices such as sensors and actuators of processes and machines. The task of this level is to transfer the information between these devices and technical process elements such as PLCs. The information transfer can be digital, analog or hybrid. The measured values may stay for longer periods or over a short period.
- In order to provide field level communication, 4-20 mA current loop, serial point-to-point communication methods are widely used. These networks consist of parallel, multiwire cables as transmission medium. The common serial communication protocol standards used in this level include RS232, RS422 and RS485. There are many other field level communication networks available which are characterized different factors such as response time, message size, etc.
- Nowadays, fieldbus technology is the most sophisticated communication network used in field level as it facilitates distributed control among various smart field devices and controller. This is a bi-directional communication system in which many variables are taken care by single transmission. Different types of fieldbuses include HART, ControlNet, DeviceNet, CAN Bus, Profibus, and Foundation Field Bus.



Introduction

Control level

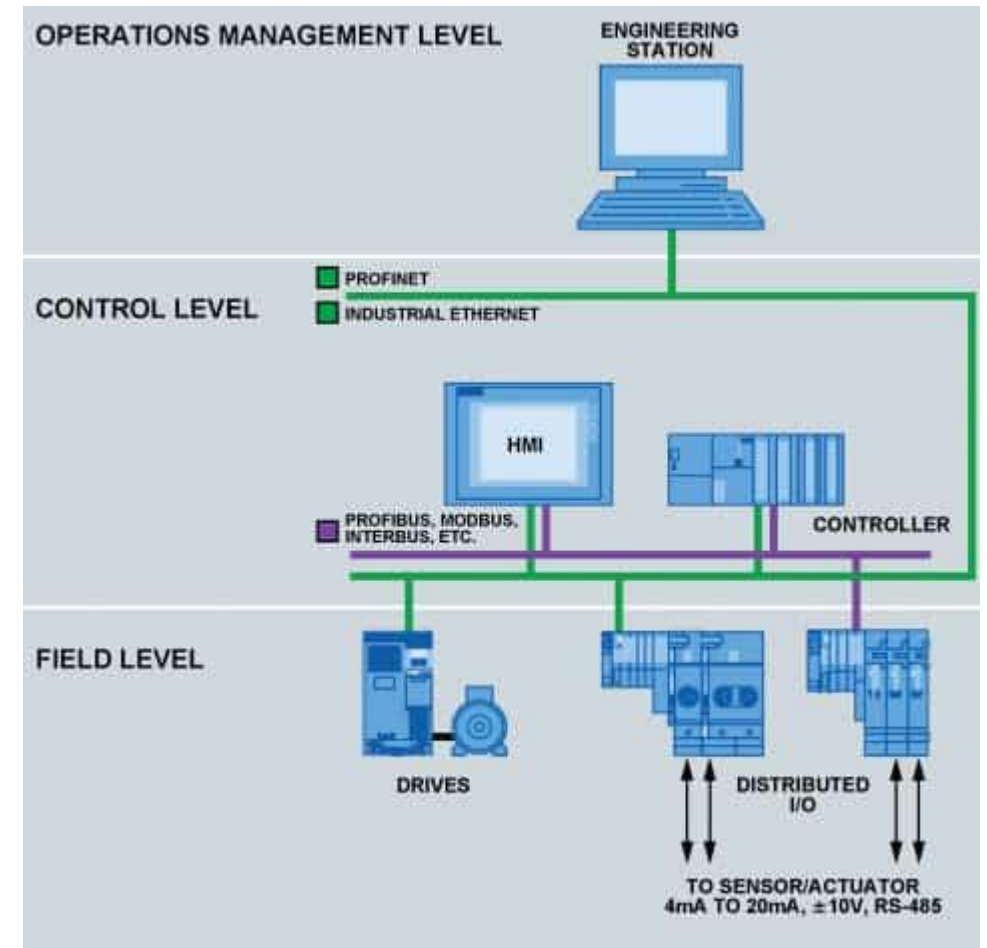
- This level consists of industrial controllers such as PLCs, distributed control units, and computer systems. The tasks of this level include configuring automation devices, loading of program data and process variables data, adjusting set variables, supervising control, displaying variables data on HMIs, historical archiving, etc.
- This level requires characteristics like short response time, high speed transmission, short data lengths, machine synchronization, constant use of critical data, etc.
- Local Area Networks (LANs) are widely used as communication networks in this level to achieve desired characteristics. The Ethernet with TCP/IP protocol is mostly used as control level network to connect control units with computers. In addition, this network acts as a control bus to coordinate and synchronize between various controller units. Some fieldbuses are also used in this level as control buses such as Profibus and ControlNet.



Introduction

Information/management level

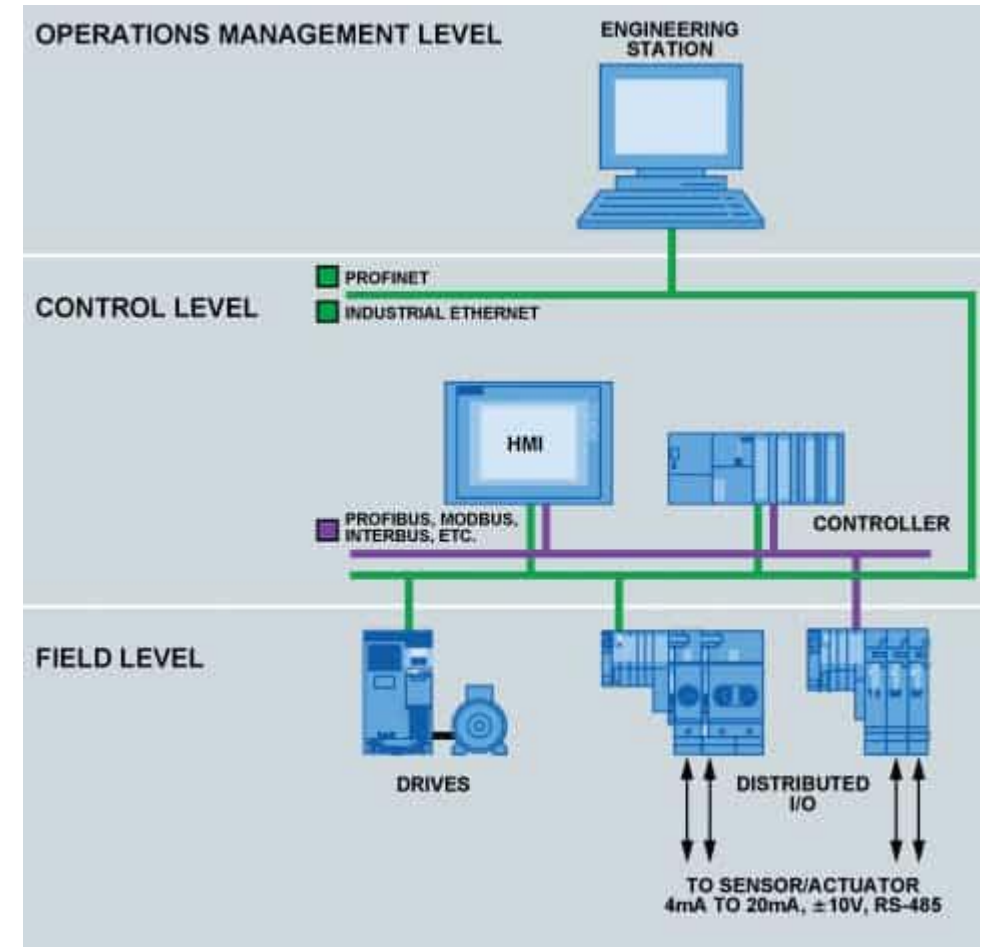
- This is the top level of the industrial automation system which gathers the information from its lower level i.e., control level. It deals with large volumes of data that are neither in constant use or time critical. Large scale networks are exists in this level.
- Ethernet WANs are commonly used as information level networks for factory planning and management information exchange. Sometimes these networks may connect to other industrial networks via gateways.



Introduction

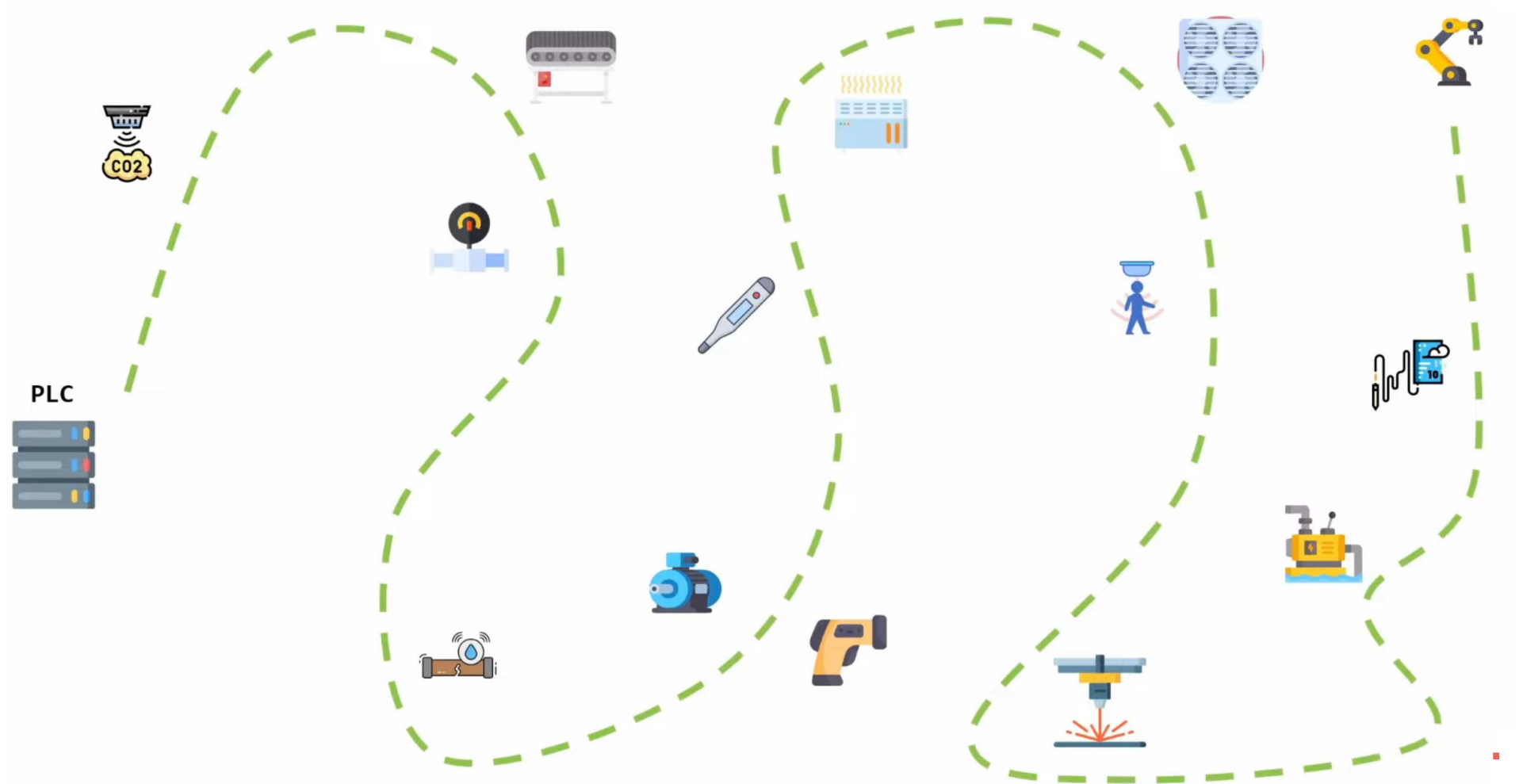
Commonly used industrial networks

- There exist many different communication networks designed to interconnect industrial field devices and various I/O modules. These are described based on certain protocols. A protocol is a set of rules that are used in communication between two or more devices. Based on these protocols, communication networks are classified into many types. Some common and popular industrial communication standards are described below
 - Serial communication (RS232, RS422, RS485)
 - HART
 - DeviceNet
 - Ethernet/ip
 - ControlNet
 - Modbus
 - CANbus / CANopen
 - Profibus
 - Profinet
 - SERCOS
 - EtherCAT
 - Etc.



Input & Outputs (I/O)

- What is I/O?
 - Sensors
 - Actuators
- Fieldbus



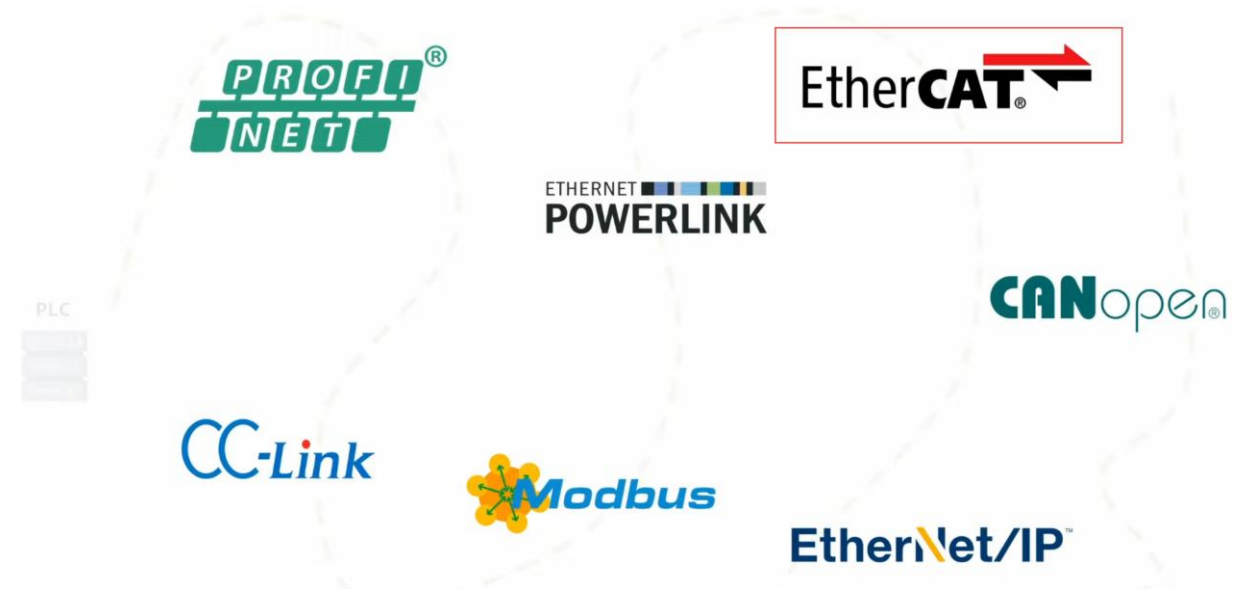
Fieldbus

- Fieldbus is an interface that connects the PLC to all these sensors and actuators
- It's a name for an industrial computer network used for real-time distributed control
- It can be implemented in a wide variety of ways and there are many different ones
 - Profinet
 - CC-Link
 - Modbus
 - Powerlink
 - EtherCAT
 - EtherNet/IP
 - CanOpen

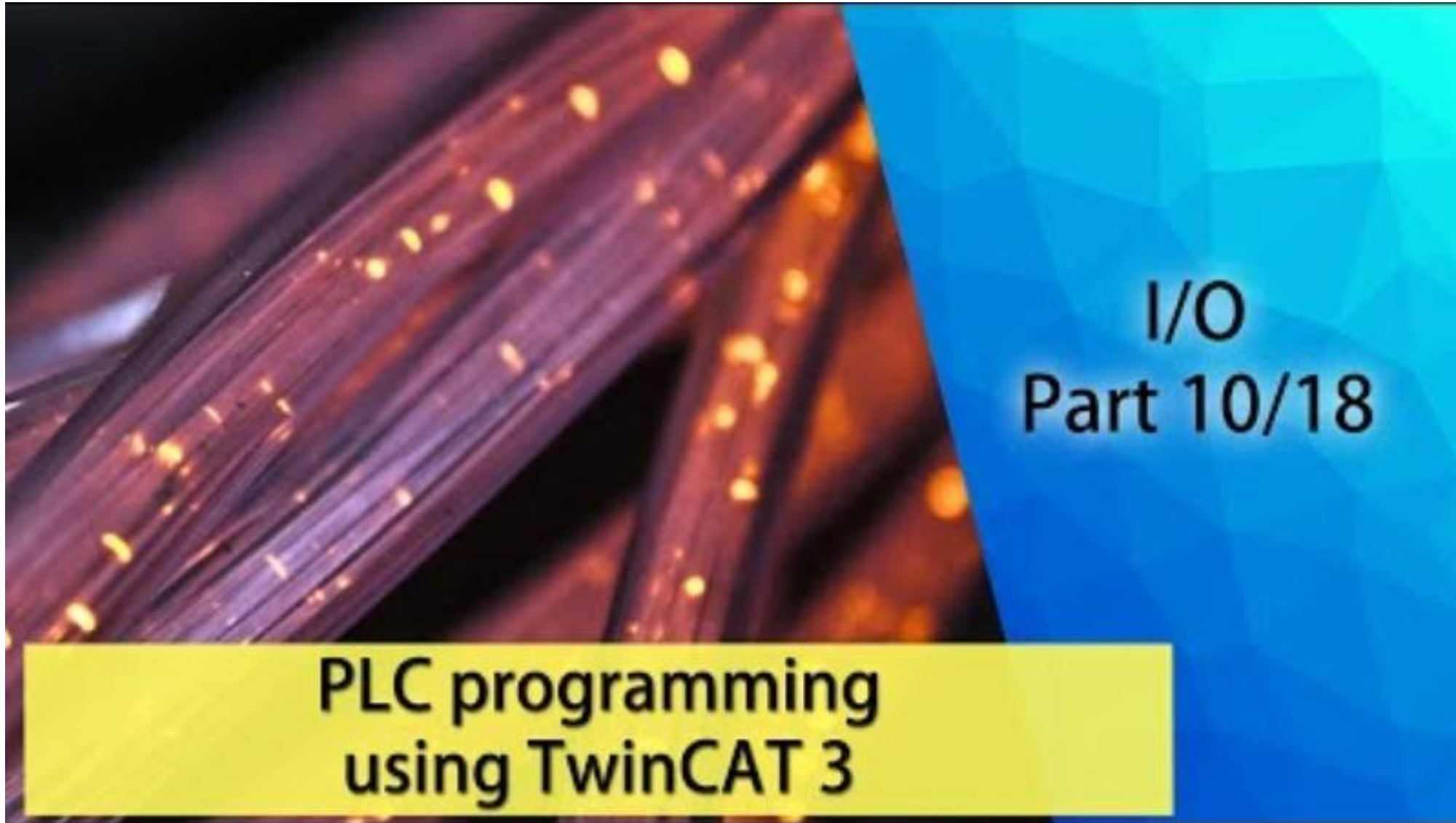


EtherCAT

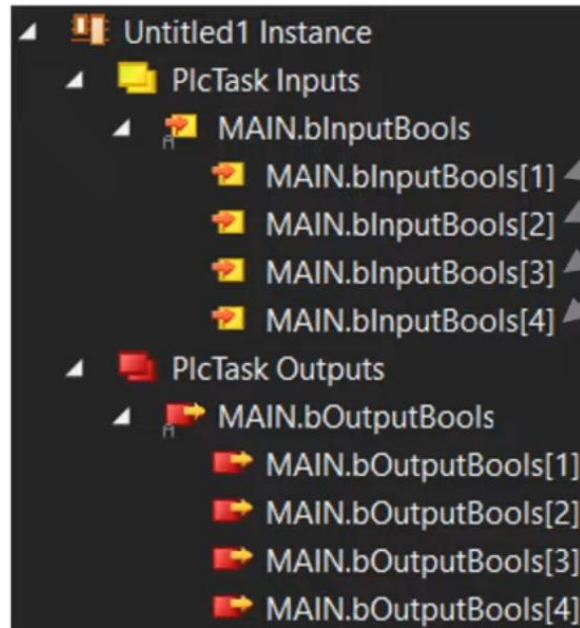
- **EtherCAT** is the fieldbus in Beckhoff PLCs
- It was invented by Beckhoff, and the real-time drivers for it are per default included in every Beckhoff PLC
- However, Beckhoff PLCs support most other Fieldbuses



EtherCAT



I/O mapping



VAR

```
bInputBools AT %I* : ARRAY[1..4] OF BOOL;  
bDoorIsOpen : BOOL;
```

END_VAR

```
bDoorIsOpen := bInputBools[3];
```

VAR

```
bOutputBools AT %Q* : ARRAY[1..4] OF BOOL;  
bOpenRelay : BOOL;
```

END_VAR

```
bOutputBools[2] := bOpenRelay;
```


Part II: ROS 2 Interface

1. Test setup
2. ADS
3. ROS 2 – ADS interface setup

Test setup

TwinCAT XAE

ROS2

TwinCAT XAR

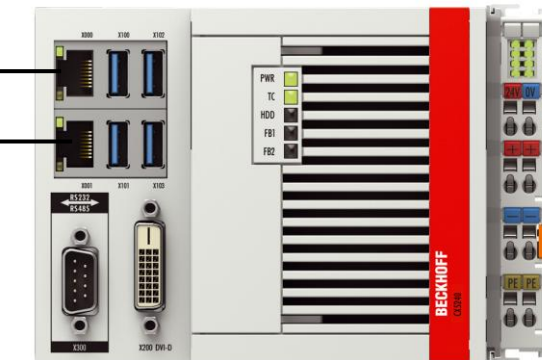


ADS client



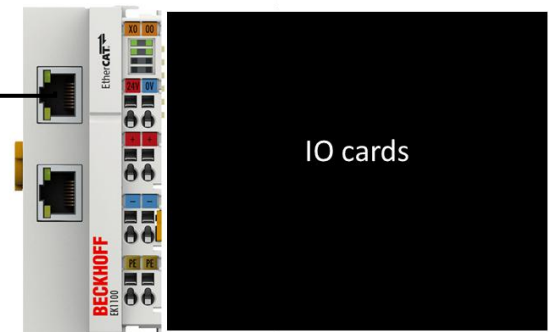
TCP/IP (ADS)

Switch



EtherCAT

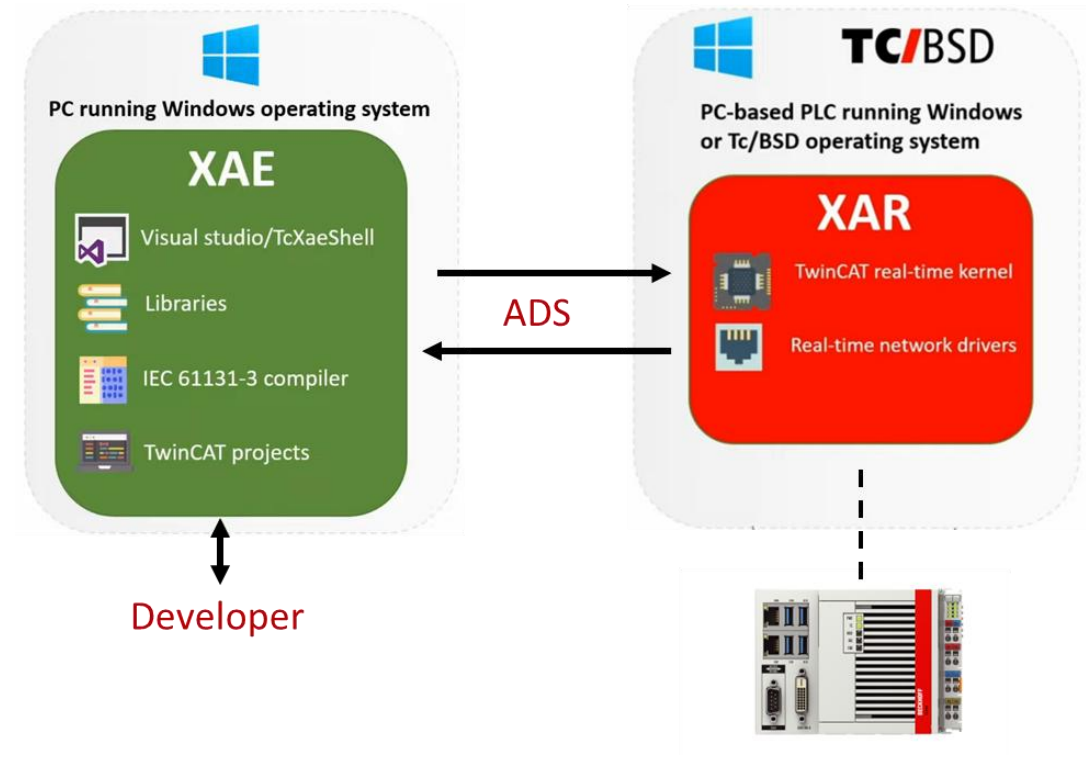
RIO



ADS

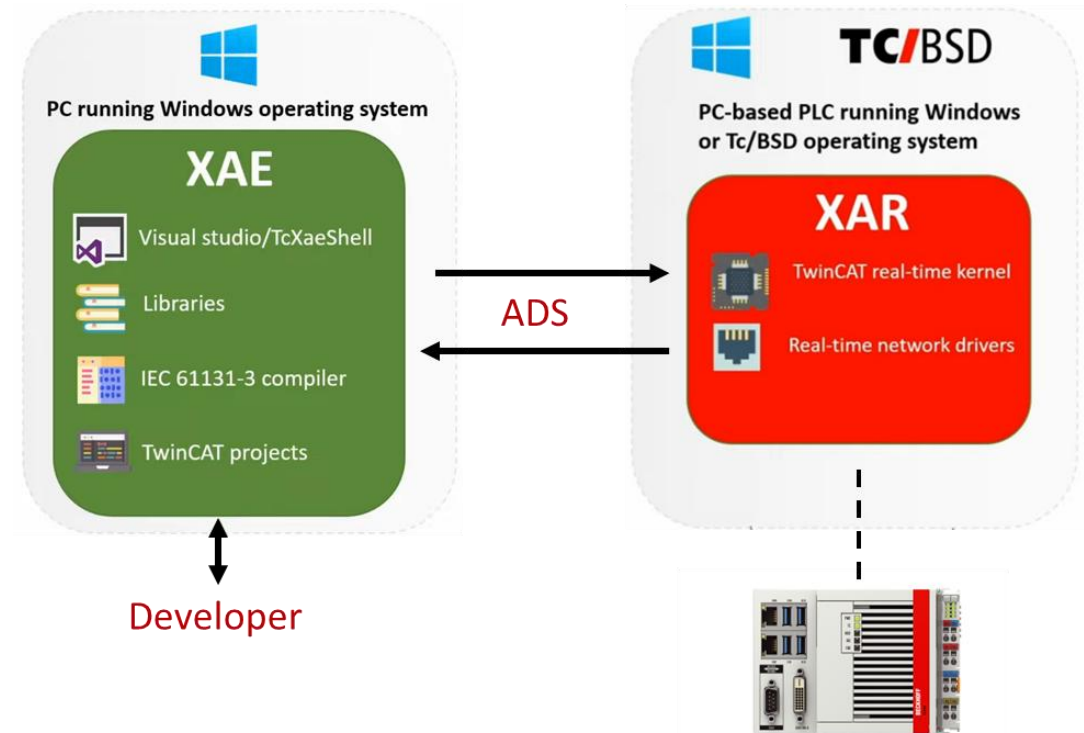
Automation Device Specification (ADS) communication protocol

- ADS is the communication protocol of TwinCAT
- It enables the data exchange and the control of TwinCAT systems
- ADS is media-independent and can communicate via serial or network connections
- ADS enables:
 - access to the process image
 - consistent data exchange
 - access to I/O tasks
 - detection of status changes
 - read-out of the PLC symbol information
 - access by variable name
 - sum commands
 - synchronous and asynchronous access
 - cyclic and event-based messages



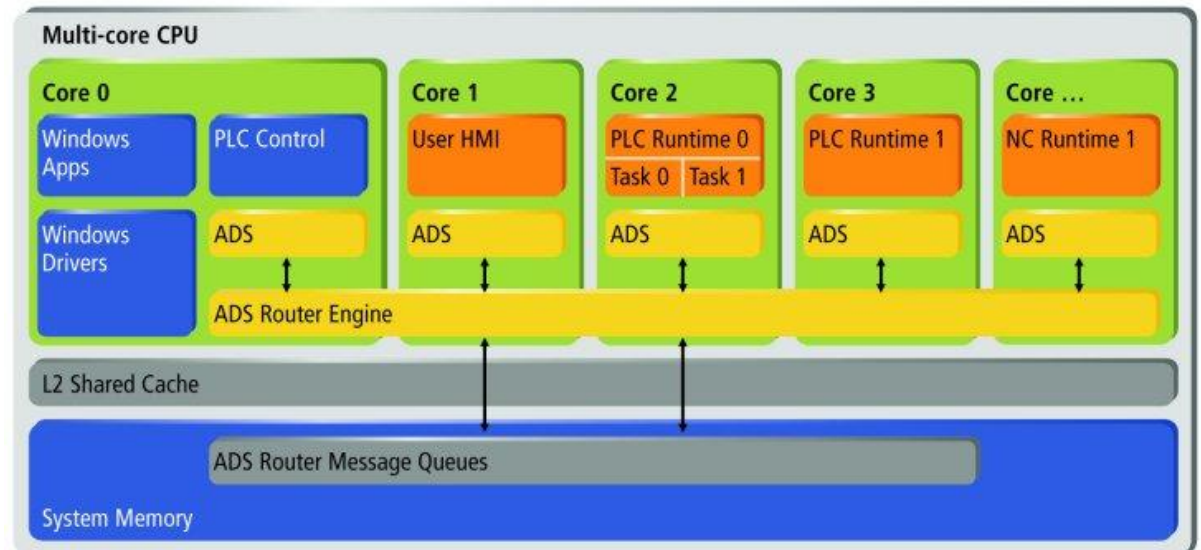
ADS

- The TwinCAT system architecture allows the individual modules of the software (e.g. TwinCAT PLC, User HMI, ...) to be treated as independent devices
- For every task there is a software module ("Server" or "Client")
- The servers in the system are the executing working "devices" in the form of software, whose operating behavior is exactly like that of a hardware device
- For this reason, we can speak of "virtual" devices implemented in the software
- The "clients" are programs which request the services of the "servers", e.g., a visualization, or even a "programming device" in the form of a program
- It is thus possible for TwinCAT to grow, since there can always be new servers and clients for tasks such as camshaft controllers, oscilloscopes, PID controllers etc.



ADS

- The messages between these objects are exchanged through a consistent ADS interface by the "message router". This manages and distributes all the messages in the system and over the TCP/IP connections
- TwinCAT message routers exist on every TwinCAT device
- This allows all TwinCAT server and client programs to exchange commands and data, to send messages, transfer status information, etc..



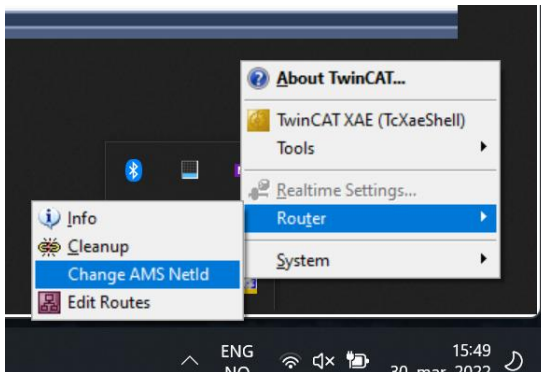
ADS

ADS device identification

- The unique identification of ADS devices is implemented with the aid of two identifiers:
 - PortNr

850	Ring 0 TC3 PLC
851	TC3 PLC Runtime System 1
852	TC3 PLC Runtime System 2
853	TC3 PLC Runtime System 3
854 - ...	TC3 PLC Runtime System 4 - ...

- NetId (e.g., **192.168.0.1.1.1**)

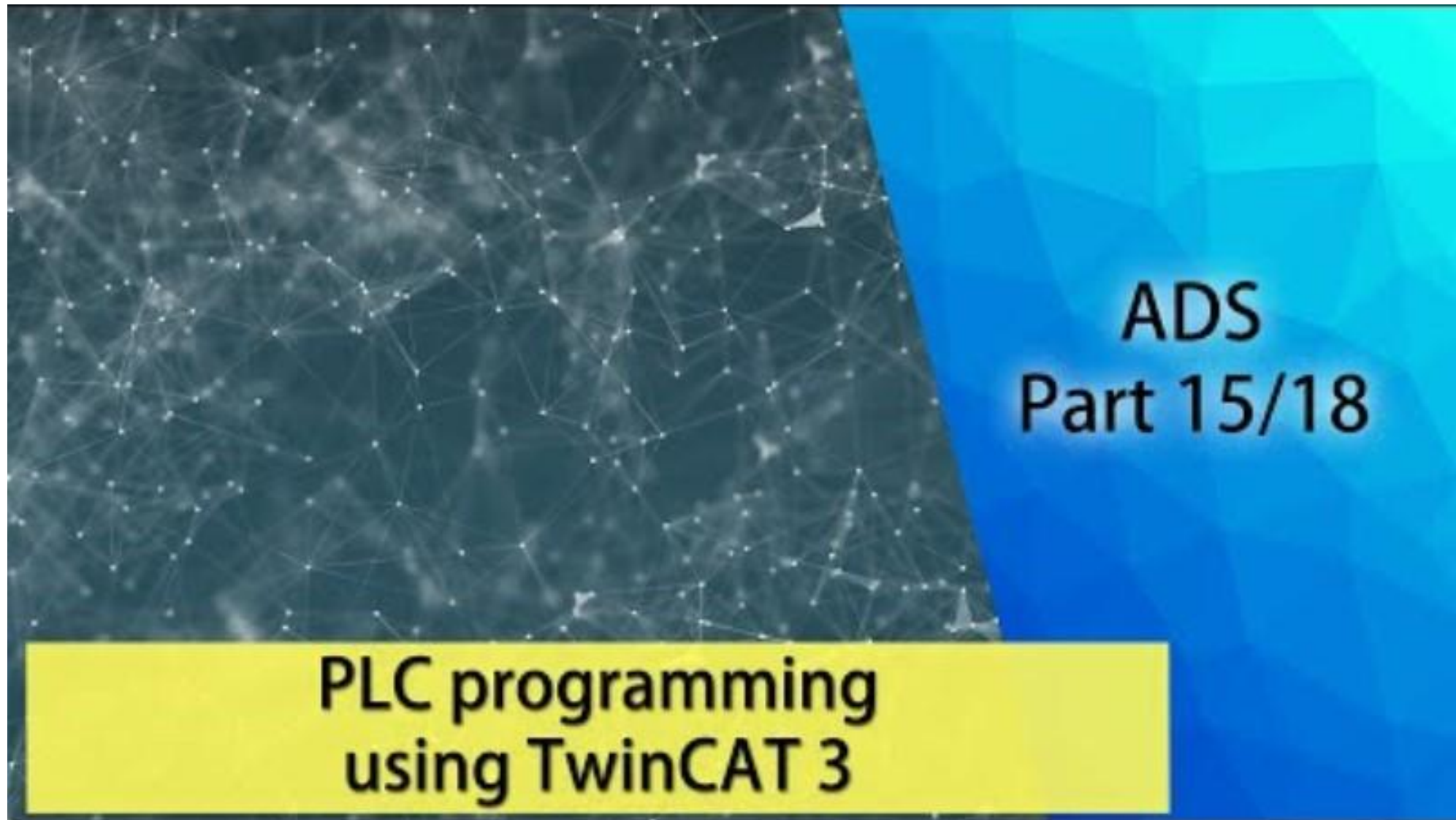


ADS

ADS commands

Command	Description
ADS Read Device Info	Reads the name and the version number of the ADS device.
ADS Read	With <i>ADS Read</i> data can be read from an ADS device
ADS Write	With <i>ADS Write</i> data can be written to an ADS device.
ADS Read State	Reads the ADS status and the device status of an ADS device.
ADS Write Control	Changes the ADS status and the device status of an ADS device.
ADS Add Device Notification	A notification is created in an ADS device.
ADS Delete Device Notification	One before defined notification is deleted in an ADS device.
ADS Device Notification	Data will carry forward independently from an ADS device to a Client
ADS Read Write	With <i>ADS ReadWrite</i> data will be written to an ADS device. Additionally, data can be read from the ADS device.

ADS (self-study)



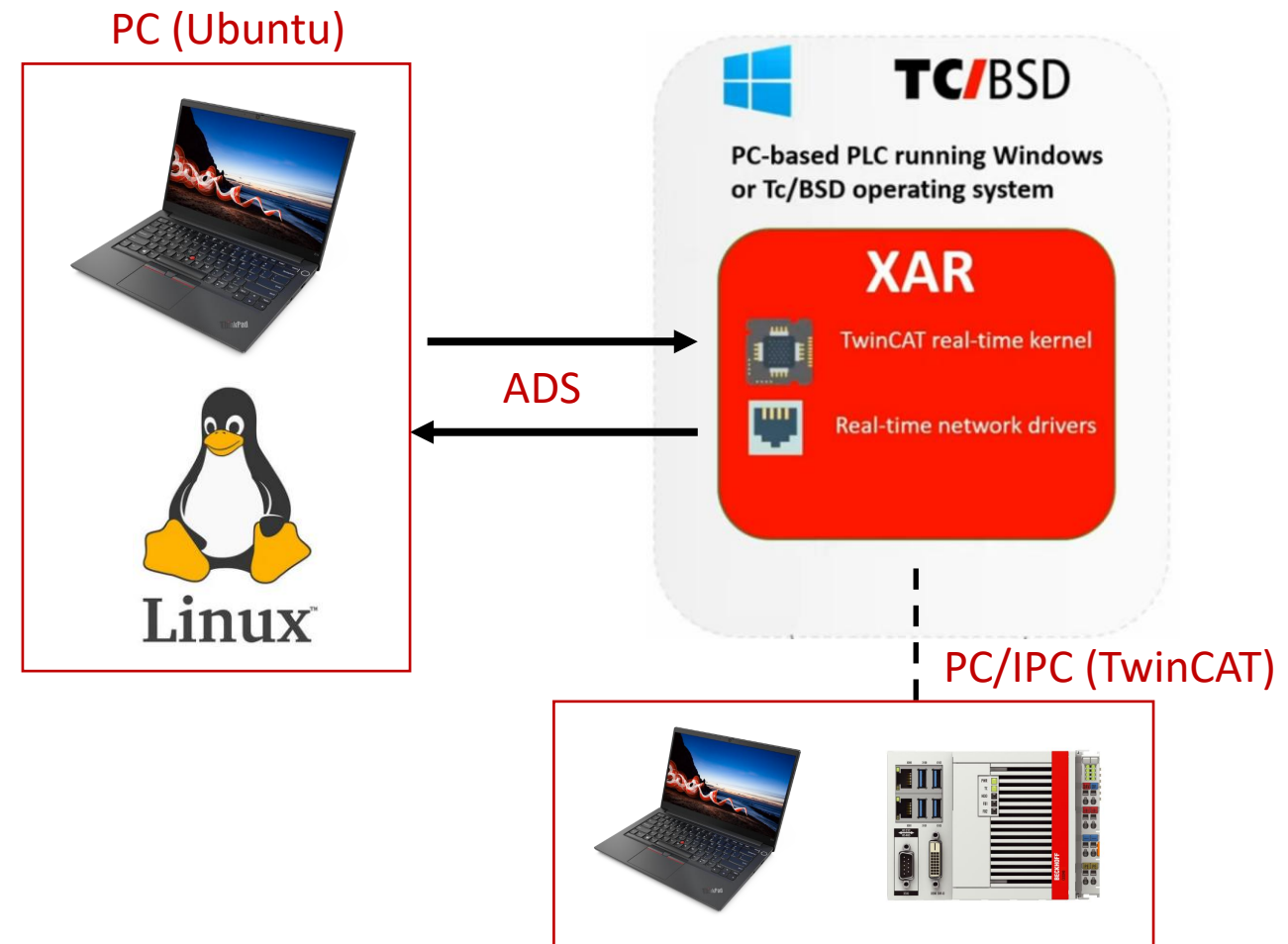
ROS 2 – ADS interface setup

Self-study:

ADS practice and interface with Linux

PLC programming using TwinCAT 3 – ADS

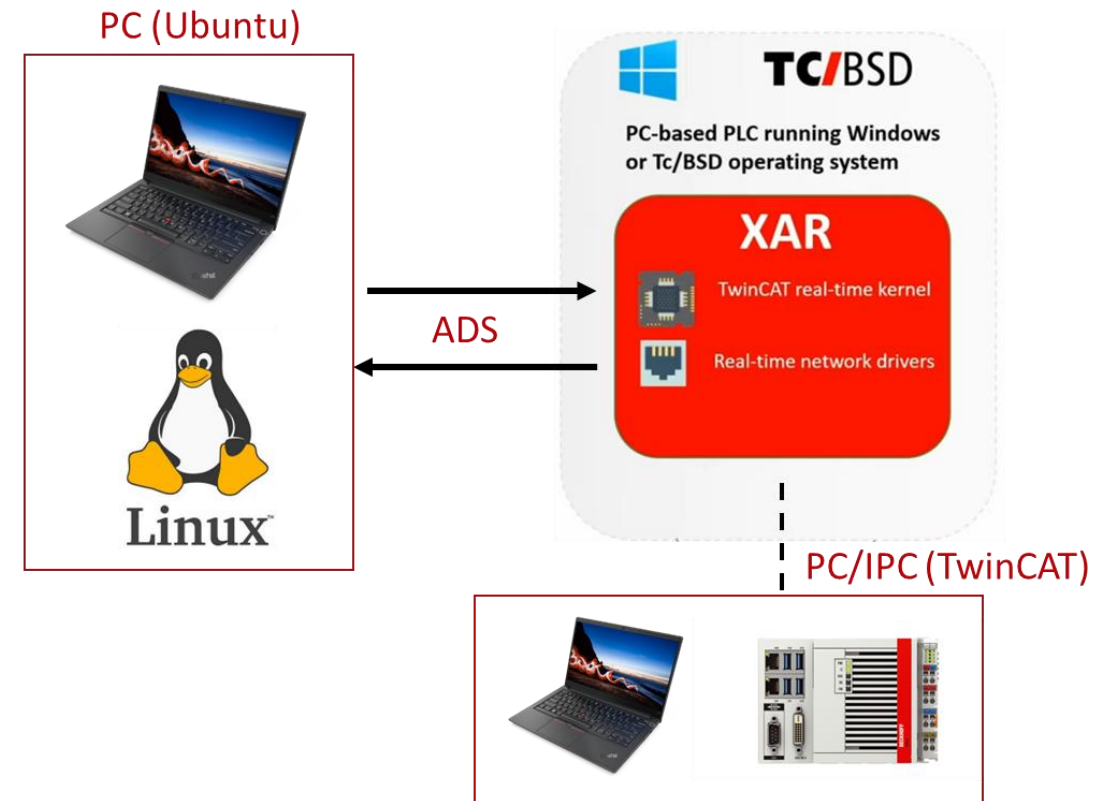
- [pyads - Python package](#)
- [PyADS for TwinCat Guide](#)
- [Connecting I/O to Python using pyads and a CX7000](#)
 - <https://github.com/benhar-dev/pyads-cx7000-as-io>



ROS 2 – ADS interface setup

TwinCAT: [ROS2-ADS-TwinCAT_Guide.pdf](#):
[MAS418-G 25H Programming for intelligent robotikk og industrielle systemer](#)

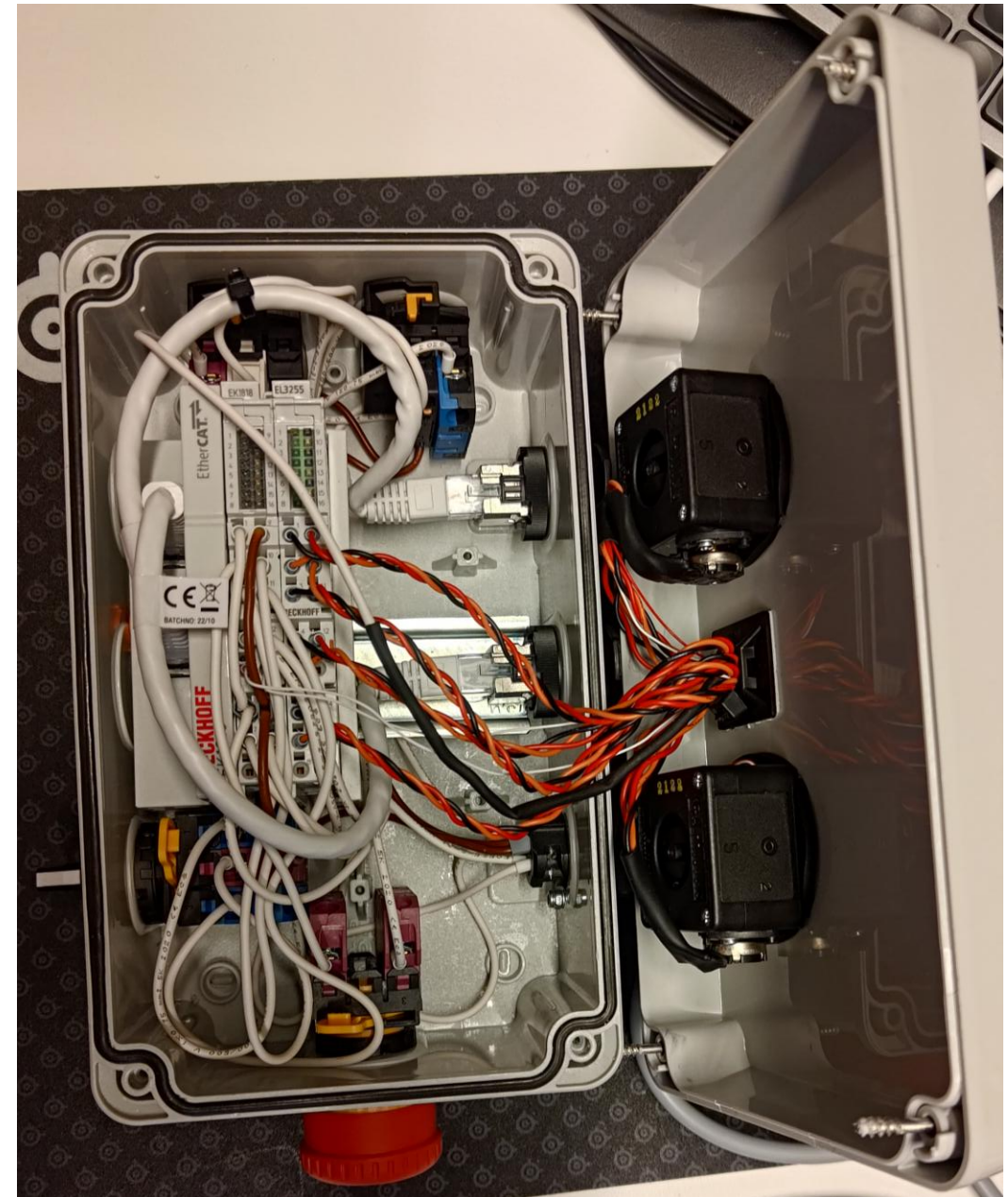
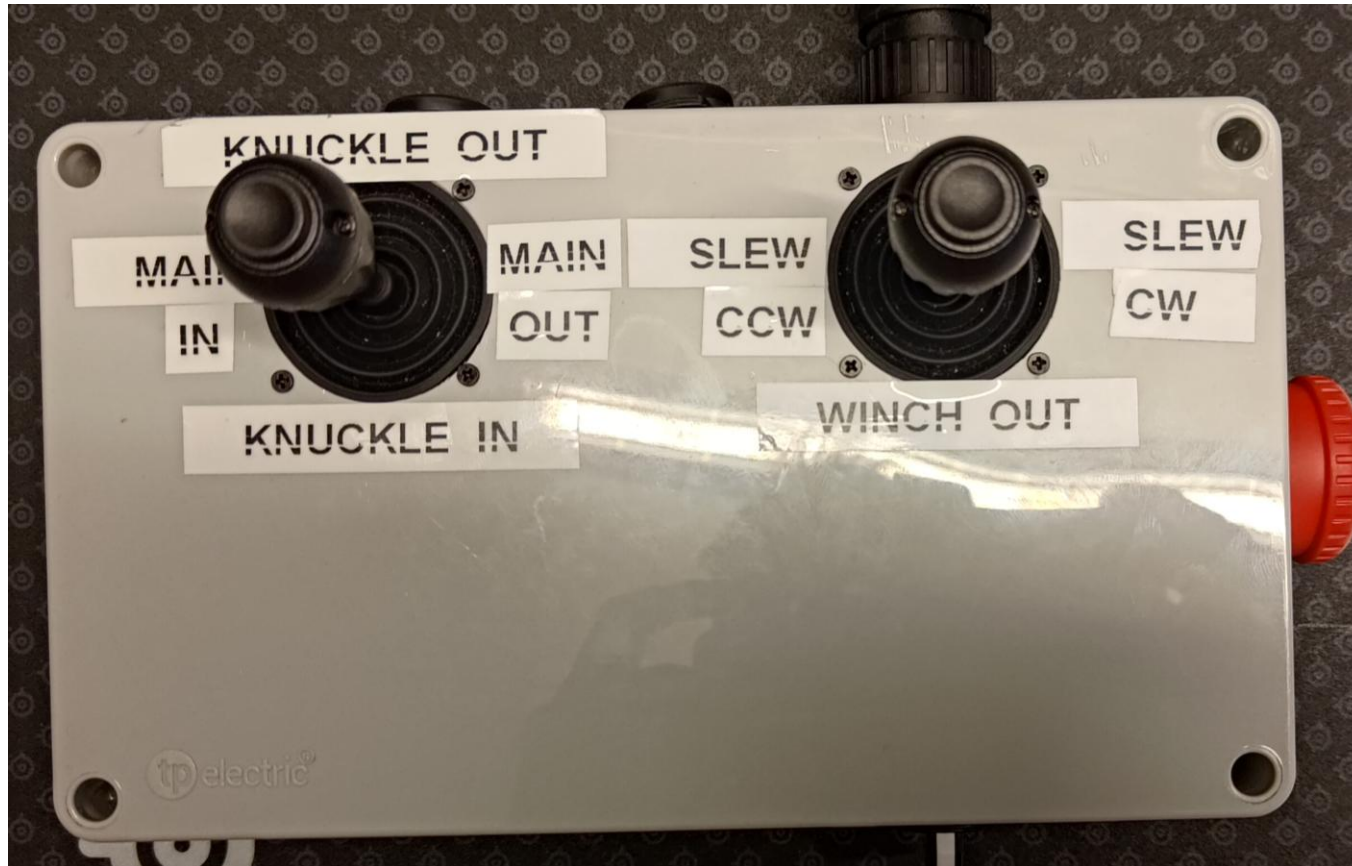
ROS 2: [kristianmk/ros2-examples-ads: Example ROS2 package for communicating with Beckhoff PLC's using ADS.](#)



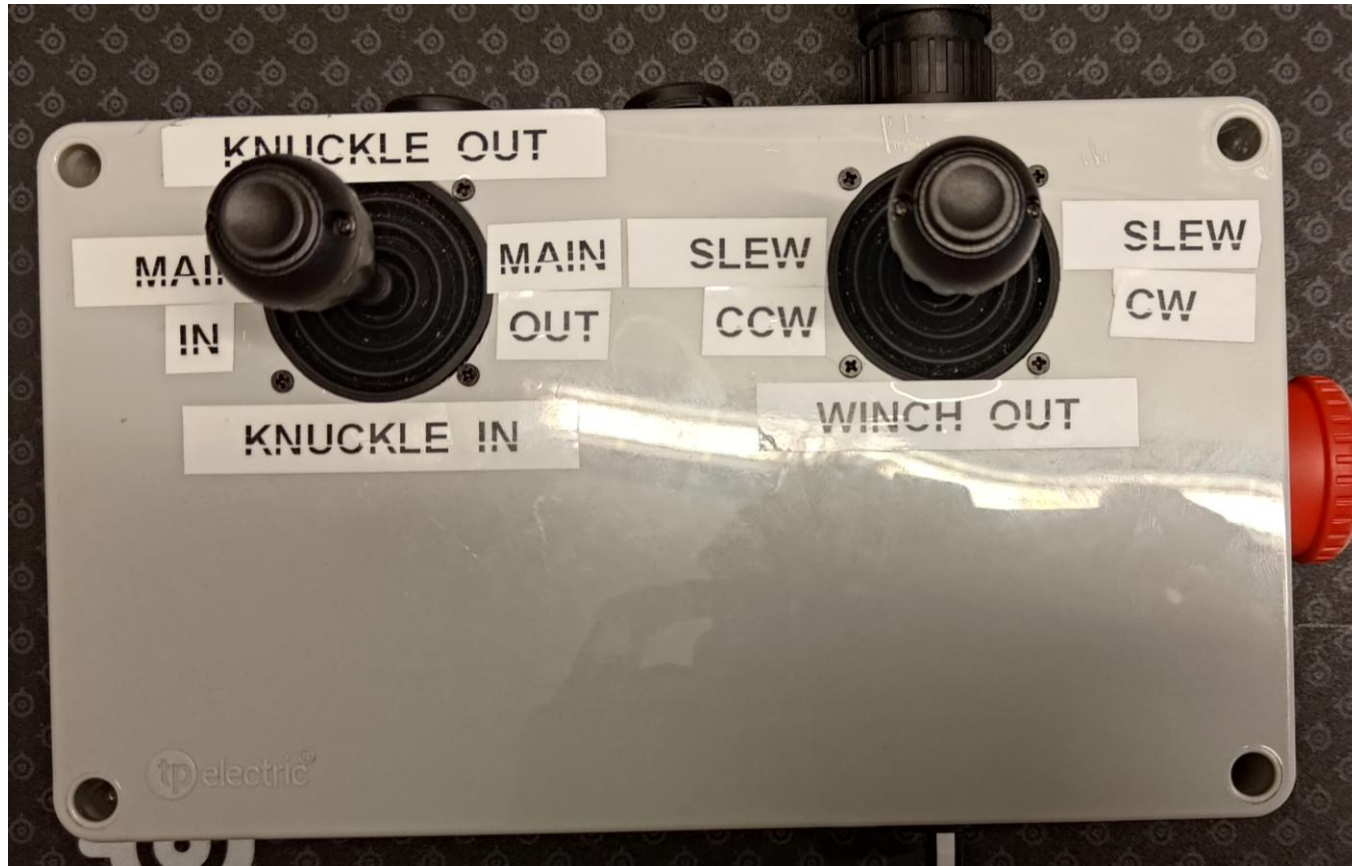
Part III: Demo

1. I/O configuration
2. I/O mapping
3. Measurement (YT Scope)

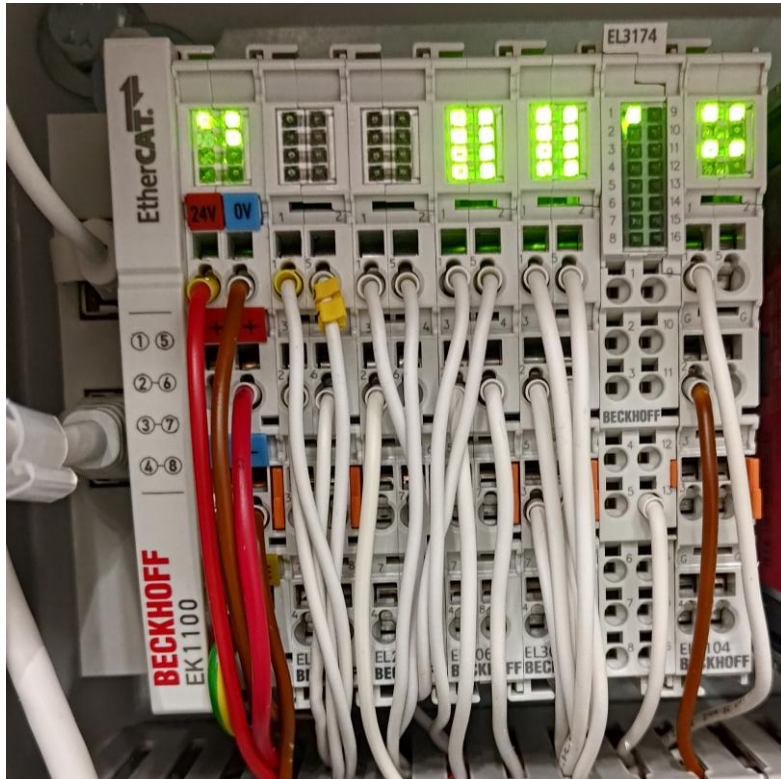
Demo



Demo



I/O



Model	Description
EL1008	8-channel digital input
EL2008	8-channel digital output
EL3068	8-channel analog input, voltage, 0...10 V
EL3068	8-channel analog input, voltage, 0...10 V
EL3174	4-channel analog input, multi-function, ± 10 V, ± 20 mA
EL4104	4-channel analog output, voltage, 0...10 V
EL9011	Bus end cover

I/O

Function	I/O
Auto	DI1:1
Manual	DI1:2
Start	DI1:3
Stop	DI1:4

Function	I/O
Green LED	DO1:1
Red LED	DO1:2
Enable Valve	DO1:3

Function	I/O
JoyX	AI1:1
JoyY	AI1:2
Xc	AI1:3
Xspool	AI1:4

Function	I/O
pS	AI2:1
pR	AI2:2
pC	AI2:3
pp	AI2:4
pA	AI2:5
pr	AI2:6

Function	I/O
Qr	AI3:1

Function	I/O
ValveOut	AO1:1

Summary

Summary

I. Machine Interface

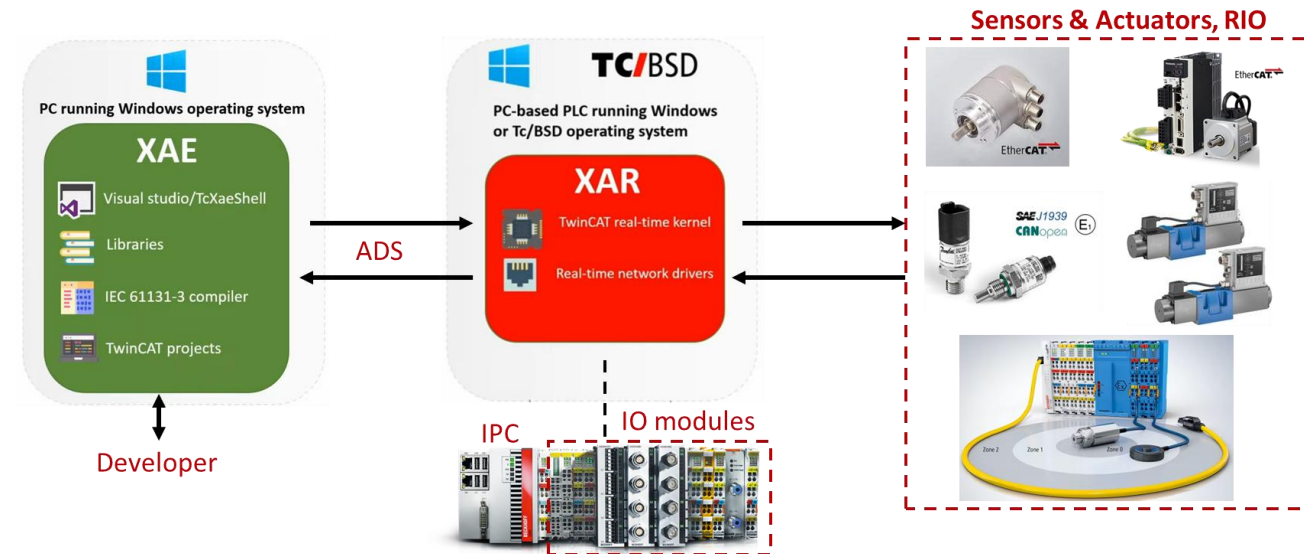
- Input & Outputs (I/O)
- Fieldbus
- EtherCAT
- ADS
- I/O mapping

II. ROS 2 Interface

- Test setup
- ADS
- ROS 2 – ADS interface setup

III. Demo

- I/O configuration
- I/O mapping
- Measurement (YT Scope)



Next Lecture

Advanced Topics

- **Homework:**
 - Work with the exercise.
 - Look at [exam from spring 2024](#) with respect to the grading method presented in [Lecture #2.1](#) slide 10.

Old exams

 MAS418 Exam S24.pdf

 MAS418_Exam_S24.tnzip

 Simulator_MAS418_Exam_S24.slx

Juli 2025								August 2025								September 2025							
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31	28	29	30	31				35	25	26	27	28	29	30	31	40	29	30					

Oktober 2025								November 2025								Desember 2025							
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40			1	2	3	4	5	44						1	2	49	1	2	3	4	5	6	7
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44	27	28	29	30	31			48	24	25	26	27	28	29	30		29	30	31				

Self-study	Part #1	Part #2	Part #3 (project)	Part Exam
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Lab exercise

#2.4 - Machine Interface

Lab exercises

-  #2.1 - Basic PLC programming
-  #2.2 - Procedural-oriented PLC programming
-  #2.3 - Object-oriented PLC programming
-  #2.4 - Machine Interface